

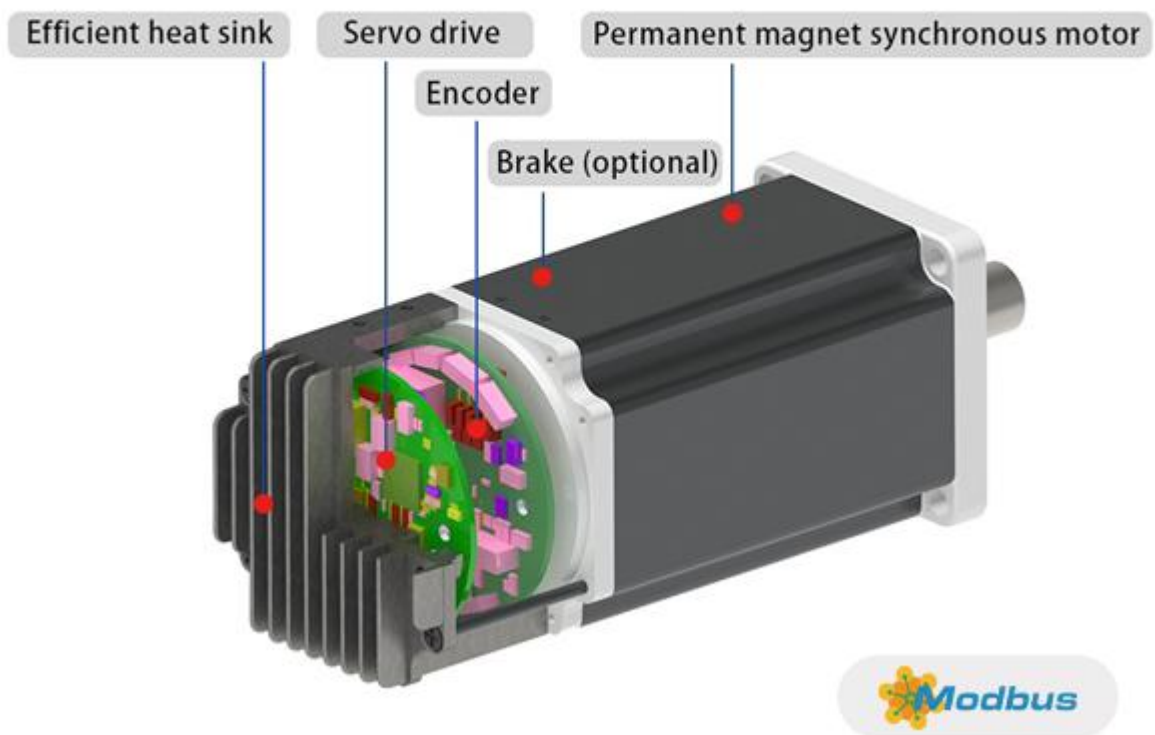


Integrated Low-voltage Servo Motor

Modbus Communication User Manual

Version number: A

INTEGRATED SERVO MOTOR (MOTOR + DRIVE)



Beijing NiMotion Control Technology Co., Ltd

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1 About the manual

1.1 Introduction

This manual explains how to program and operate the integrated low-voltage Servo motor manufactured by Beijing NiMotion Control Technology Co., Ltd.

1.2 Meaning of Numerical Values

In this manual, numerical values are usually in decimal notation. If they have to be in hexadecimal notation, the subscript "h" at the end of the number is used, for example: 1003_h.

Directory index and sub-index of object are represented as follows:

<index>:<sub-index>

For example, sub-index 02 of object 1003_h is represented as **1003_h:02_h(0004_h)**.

1.3 Bit

Bits of an object always start from 0 at LSB.

	MSB				LSB				
Bit Number	7	6	5	4	3	2	1	0	
Bits	0	1	0	1	0	1	0	1	55 _{hex} = 85 _{dec}

2 Communications Network

2.1 Modbus Overview

The communication mode of the Modbus protocol is of the single-master/multi-slave type. Only the master station can send out queries (inquiries). The slave stations carry out the processing as required by the queries and reply with response messages.

2.2 Modbus-RTU Transmission Mode

When the device uses the RTU (Remote Terminal Unit) mode for Modbus serial link communication, each 8-bit byte in the message contains two 4-bit hexadecimal characters. The main advantage of this mode is a higher data density, with a higher throughput at the same baud rate compared to ASCII mode. Each message must be sent as a continuous stream of characters.

Frame Description:

Table 2- 1

Child Node Address	Function Code	Digital	CRC
1 byte	1 byte	0 to 252 bytes	2 bytes CRC low CRC high

The message frame is distinguished by an idle interval of at least 3.5 character time, called t3.5, and this module adopts the standard t3.5 as the idle interval distinction. the RTU frame transmission idle interval distinction diagram is shown in Figure 1.

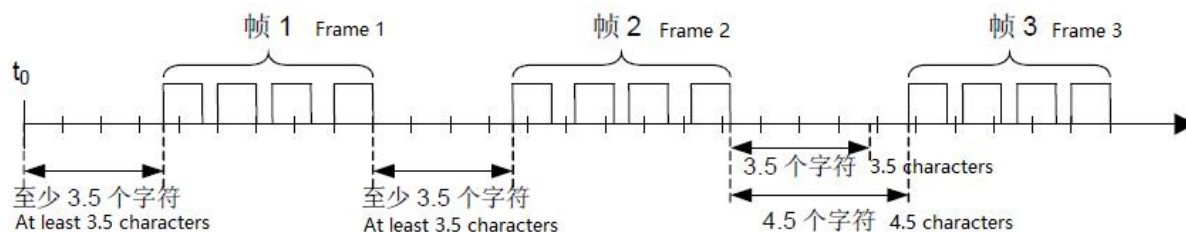


Figure 2- 1 Schematic diagram of RTU frame transmission idle interval differentiation

2.3 Data Type

Modbus is based on a series of data models on tables with different characteristics. The two basic tables are:

Table 2- 2

Basic Forms	Object Type	Type of Access	Element
Input register	16-bit	read-only	The I/O system provides this type of data
Holding register	16-bit	Read and write	Changing this type of data through the application

2.4 Modbus-RTU Function Code

Table 2- 3

	Function Code Name	Function Code	Sub-function Code	Broadcast Message
16-bit access	Read Input Register	0x04	-	unsupported
	Read Holding Register	0x03	-	unsupported
	Write Single Register	0x06	-	supported
	Write multiple registers	0x10	-	supported

2.4.1 Read Holding Register (0x03)

Read the current value of the baud rate 200C_h : 03_h (0231_h).

request message

Table 2- 4

Slave Address	Function Code	Register Address	Number of Registers	CRC Value
01	0x03	02 31	00 01	-

response message

Table 2- 5

Slave Address	Function Code	Byte Count	Register Value	CRC Value
01	0x03	02	00 00	-

The current baud rate value is 0 (9.6kbps).

2.4.2 Read Input Register (0x04)

Read the value of input voltage 200B_h : 15_h (01F7_h).

request message

Table 2- 6

Slave Address	Function Code	Register Address	Number of Registers	CRC Value
01	0x04	01 F7	00 01	-

response message

Table 2- 7

Slave Address	Function Code	Byte Count	Register Value	CRC Value
01	0x04	02	00 E7	-

The current input voltage value is 0x00E7 (23.1V).

2.4.3 Write Single Holding Register (0x06)

Set the baud rate 200C_h: 03_h (0231_h) of slave address 0x01 to 0x01 (9.6kbps).

request message

Table 2- 8

Slave Address	Function Code	Register Address	Register Value	CRC Value
01	0x06	02 31	00 01	-

response message

Table 2- 9

Slave Address	Function Code	Byte Count	Register Value	CRC Value
01	0x06	02 31	00 01	-

2.4.4 Write Multiple Holding Registers (0x10)

Set the slave address 200C_h: 02_h (0230_h) of slave address 0x01 to 0x02 and the baud rate 200C_h: 03_h (0231_h) to 0x01(9.6kbps).

request message

Table 2- 10

Slave Address	Function Code	Register Address	Number of Registers	Byte Count	Register Value	CRC Value
01	0x10	02 30	00 02	04	00 02 00 01	-

response message

Table 2- 11

Slave Address	Function Code	Register Address	Number of Registers	CRC Value
01	0x10	02 30	00 02	-

2.5 Communication Exception Response

When a master device sends a request to a slave device, the master expects a normal response. The master's query may result in one of the following four events:

- If the slave device receives the request without communication errors and can handle the query normally, then the slave device will return a normal response.
- If the slave does not receive the request due to a communication error, the response cannot be returned. The master program will process it as a timeout.
- If the slave receives a request but detects a communication error (parity, CRC...), then no response can be returned. The master program will be processed on timeout.
- If the slave device receives a request for no communication error, but cannot process the request (e.g., if requesting a read of a non-existent input register), then the slave will return an exception response informing the client of the cause of the error.

The exception response message has two fields that are different from the normal response message:

Function Code Field:

In a normal response, the slave assigns the function code of the original request in the function code field of the response. All function codes have an MSB of 0 (they all have values below hex 80). In an exception response, the slave sets the MSB of the function code to 1. This makes the function code value in the exception response hex 80 higher than the function code value in the normal response.

By setting the function code MSB, the master's application program recognizes the exception response and can detect the data fields of the exception code.

Data Fields:

In a normal response, the slave can return data or statistical values (any information requested in the request) in the Data field.

In an exception response, the slave returns an exception code in the data field. This indicates the reason for the exception.

The exception response message for all supported function codes is the function code plus 0x80.

All exception function codes are 0x83, 0x84, 0x86, and 0x90.

The exception response message corresponding to function code 0x06 is as follows:

Table 2- 12

Slave Address	Function Code	Exception Code	CRC Value
01	0x86	01	-

Table 2- 13

Exception Code	Exception Name	Description
01	Illegal function code	Function code cannot be recognized, not within 0x00~0x0F.
02	Illegal data address	The data address exceeds the definition.
03	Illegal data value	Values outside of register storage
04	Slave equipment failure	Generate non-recapturable errors
05	Recognize	Slaves take longer to process, requiring frequent interrogation by the master
06	Slave equipment busy	Wait for the host to send a request while the slave device is idle
12	Slave equipment alarm	Alarm exists for current motor status.

2.6 Communication Parameter Configuration

2.6.1 Baud Rate Setting

The communication rate of the device can be changed by setting the communication baud rate register of the RS-485 communication interface, but the parameters need to be saved after setting the baud rate, and will take effect after the device is next powered on or rebooted.

The index corresponding to the baud rate parameter is 200C_h : 03_h (0231_h), and the function codes of the operable holding registers are 0x03, 0x06, and 0x10. The factory baud rate default value is 115.2Kbps.

Example of setting the communication baud rate of the slave address to 115.2 Kbps.

Request messages sent

Table 2- 14

Slave Address	Function Code	Register Address	Register Value	CRC Value
01	0x06	02 31	00 05	-

response message

Table 2- 15

Slave Address	Function Code	Register Address	Register Value	CRC Value
01	0x06	02 31	00 05	-

Saving the set communication baud rate parameter.

request messages

Table 2- 16

Slave Address	Function Code	Register Address	Number of Registers	Byte Count	Register Value	CRC Value
01	0x10	00 26	00 02	04	65 76 61 73	-

response message

Table 2- 17

Slave Address	Function Code	Register Address	Number of Registers	CRC Value
01	0x10	00 26	00 02	-

Table of setting values and correspondence with baud rate:

Table 2- 18

Hexadecimal	Baud Rate
0	9.6Kbps
1	9.6Kbps
2	19.2Kbps
3	38.4Kbps
4	57.6Kbps
5	115.2Kbps
6	256Kbps
7	500Kbps
8	1Mbps
9	1.5Mbps

2.6.2 Node ID

The address of the device can be set by setting the slave address 200C_h :02_h (0230_h), and the range of setting is 1~247. The factory default slave address is 0x01. After the successful setting of the slave address, it is necessary to save the parameter setting, and it will take effect after the device is powered on or rebooted next time.

2.6.3 Network Data Format

The network data format for device communication is changed by setting the network data format 200C_h : 04_h (0232_h). The corresponding table of setting values is shown in the table below.

Table 2- 19

hexadecimal	descriptive
0	8 data bits, even parity, 1 stop bit
1	8 data bits, odd parity, 1 stop bit
2	8 data bits, no parity, 1 stop bit
3	8 data bits, no parity, 2 stop bits

2.6.4 Connection of Communication Lines

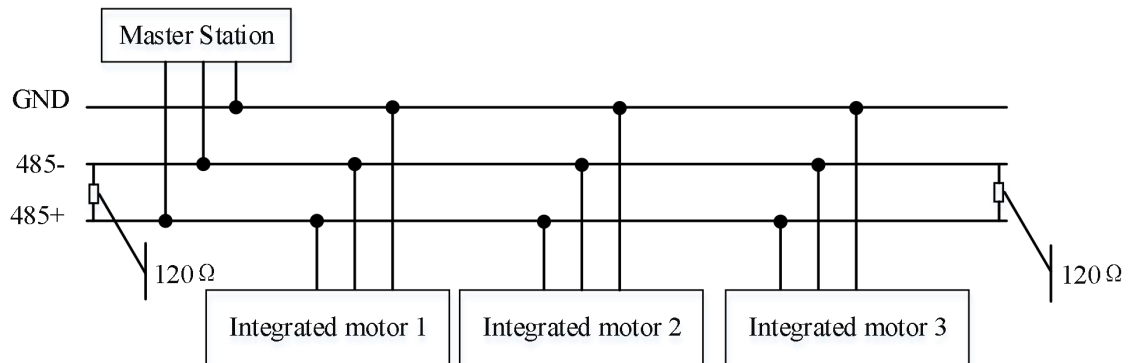


Figure 2- 2

2.6.5 Broadcast Seizure

When the communication parameters of the motor are not sure to forget, it can be restarted after power failure by broadcasting telegram to seize it. The broadcast preemption request message should be sent within 1 second before the motor is powered on, and the RUN indicator blinks after successful preemption. After successful seizure, the communication parameters of the motor are restored to the default parameters, i.e., the communication parameters are restored to the factory settings, the slave address is equal to 1, the baud rate is equal to 115200, the parity bit is none, there are 8 data bits and 1 stop bit. The motor serial number adopts the byte sequence of hexadecimal high byte in the front.

Table 2- 20 Broadcast preemption request messages

Slave Address (1B)	Function Code (1B)	Motor Serial Number (4B)	CRC Value (2B)
0x00	0xD2	0x00 00 00 00	-

3 Equipment Control

3.1 The CiA402 State Machine

The operating state of integrated low-voltage servo motor is switched by a state machine, which is defined in the CiA402 standard. The motor drive is controlled by the control word 6040_h(0380_h), and the actual state of the device can be read back from the status word object 6041_h (0381_h).

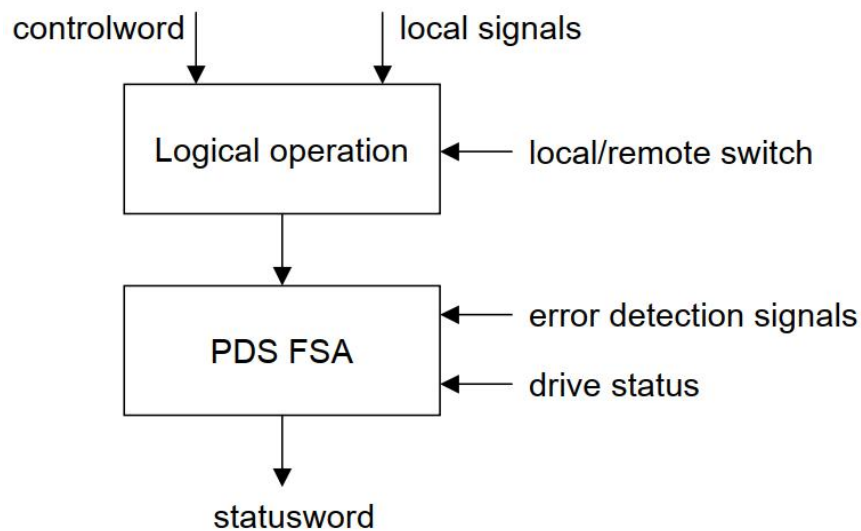


Figure 3- 1

3.1.1 The State Transition Diagram

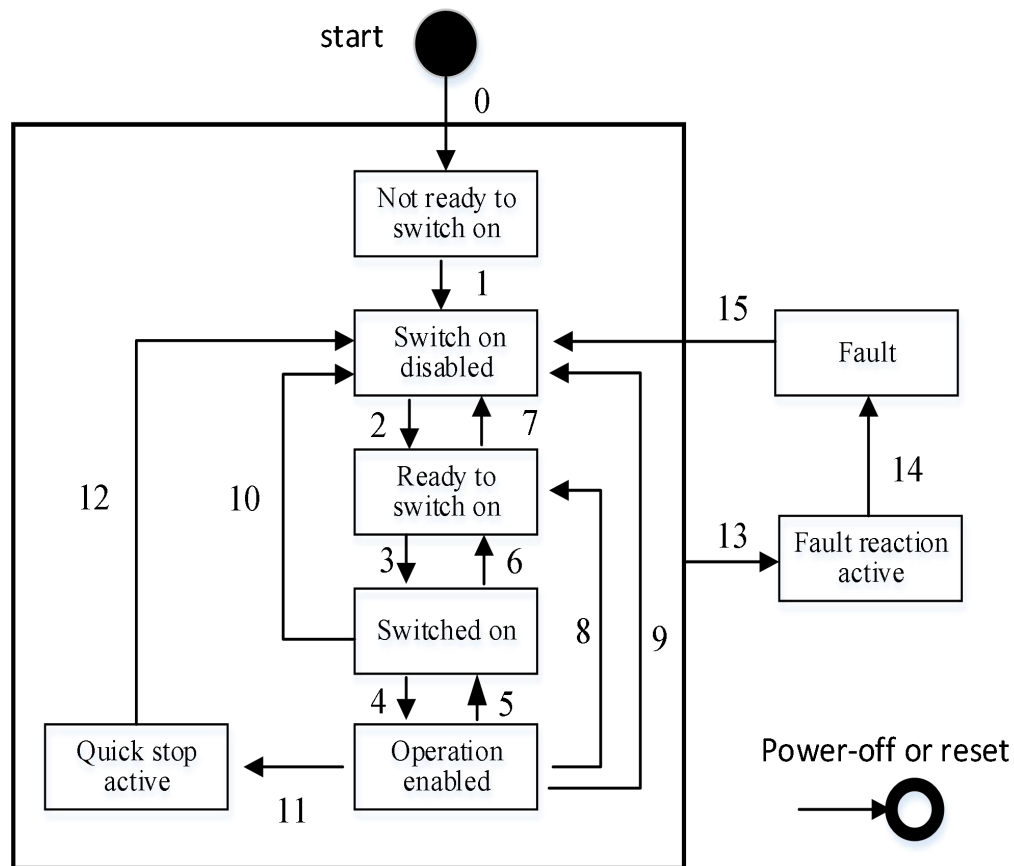


Figure 3- 2

Each state corresponds to the motor as shown in the table below:

Table 3- 1

Function	FSA states							
	Not ready to switch on	Switch on disabled	Ready to switch on	Switched on	Operation enabled	Quick stop active	Fault reaction active	Fault
Brake applied, if present	Yes	Yes	Yes	Yes	Yes/No	Yes/No	Yes/No	Yes
Low-level power applied	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
High-level power applied	Yes/No	Yes/No	Yes/No	Yes	Yes	Yes	Yes	Yes/No
Drive function enabled	No	No	No	No	Yes	Yes	Yes	No
Configuration allowed	Yes	Yes	Yes	Yes	Yes/No	Yes/No	Yes/No	Yes

The control word and state machine state switching is shown in the table below:

Table 3- 2

Command	Bits of the controlword					Transitions
	Bit7	Bit3	Bit2	Bit1	Bit0	
Shutdown	0	×	1	1	0	2,6,8
Switch on	0	0	1	1	1	3
Switch on + enable operation	0	1	1	1	1	3+4 (NOTE)
Disable voltage	0	×	×	0	×	7, 9, 10, 12
Quick stop	0	×	0	1	×	7, 10, 11
Disable operation	0	0	1	1	1	5
Enable operation	0	1	1	1	1	4, 16
Fault reset		×	×	×	×	15

NOTE: Automatic transition to Enable operation state after executing SWITCHED ON state functionality

The status words for each state are listed below:

Table 3- 3

Statusword	PDS FSA state
xxxx xxxx x0xx 0000b	Not ready to switch on
xxxx xxxx x1xx 0000b	Switch on disabled
xxxx xxxx x01x 0001b	Ready to switch on
xxxx xxxx x01x 0011b	Switched on
xxxx xxxx x01x 0111b	Operation enabled
xxxx xxxx x00x 0111b	Quick stop active
xxxx xxxx x0xx 1111b	Fault reaction active
xxxx xxxx x0xx 1000b	Fault

Note: x means that this bit does not been care, bit 8 ~ bit15 has different meaning in different modes.
Please check its specific status in each mode.

3.1.2 Control Word

The bits of the control word, defined as follows:

15	10	9	8	7	6	3	2	1	0
11					4				
ms	r	oms	h	fr	oms	eo	qs	ev	so
MSB									LSB

LEGEND: ms = manufacturer-specific; r = reserved; oms = operation mode specific; h = halt; fr = fault reset; eo = enable operation; qs = quick stop; ev = enable voltage; so = switch on enable voltage; so = switch on

● Note

It is meaningless to assign each bit of the control word individually,It must be combined with other bits to form a control instruction. Bit 0 ~bit 3 and bit 7 have the same meaning in each mode, Only by sending commands sequentially can the motor drive be guided into the predicted state according to the switching process of the

CiA402 state machine, and each command corresponds to 1 definitive state. Bit4 ~ bit6 are related to each operation mode (please check the control instructions in different modes).

3.1.3 Status Word

The bits of the status word are defined as follows:

15	13	11	10	9	8	7	6	5	4	3	2	1	0
14	12												
ms	oms	ila	tr	rm	ms	w	sod	qs	ve	f	oe	so	rtso
MSB													LSB

Table 3- 4

Bit	Description
0	ready to switch on
1	switched on
2	operation enabled
3	fault
4	voltage enabled
5	quick stop
6	switch on disabled
7	warning
8	manufacturer specific
9	remote
10	target reached
11	internal limit active
12-13	operation mode specific
14-15	manufacturer specific

● Notes:

It is meaningless to assign each bit of the status word individually, It must be combined with other control bits to form the current state feedback of the motor. In each stepping mode, bit 0-bit9 has the same meaning, by which the motor feeds back a certain status after sending commands by the control word 6040_h sequentially. Besides, bit10, bit11 and bit15 have the same meaning in each mode, by which the motor feeds back its status after a certain operation mode is executed.

3.1.4 The Stop Mode

The stop mode is the action of the state machine which leaving the "Operation enabled" state, corresponding settings are as follows:

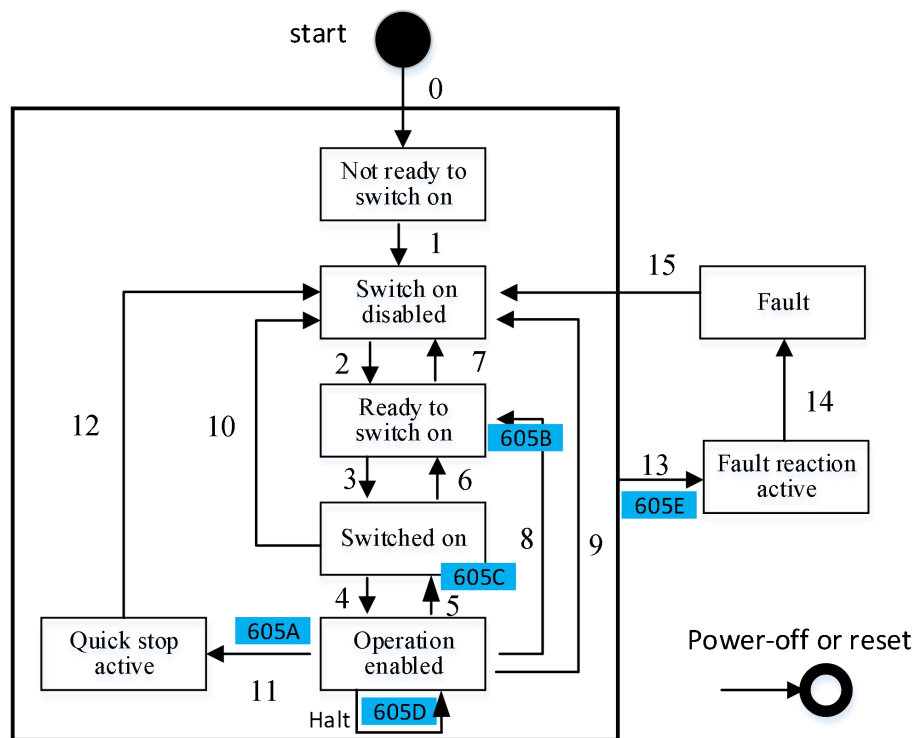


Figure 3- 3

● Fault

When a error or a warning occurs, automatically the servo switches to the Fault state which is related to 605E_h (03C1_h) (Fault response option code). When 605E_h (03C1_h) = -1, the fault reaction is related to the reaction code in the fault code which is configurable. When 605E_h (03C1_h)_h > 0, the fault reaction is selected by the object dictionary 605E_h (03C1_h). The relationship between the setting value in 605E_h (03C1_h) and the fault reaction is as follows:

Table 3- 5

Setting Value	Stop Mode
-1	The stop mode is related to fault reaction option code in alarm code. See "9.1 alarm code" for details
0x00	Slow down freely and keep operating freely.
0x01	Slow down on slow down ramp set in 6084 _h .
0x02	Slow down on quick slow down ramp set in 6085 _h .

● Quick Stop

A quick stop is performed when the control word 6040_h (0380_h) = 0x02 in a non-fault condition.

The quick stop method is selected via the object dictionary 605A_h (03BF_h) and the meaning of the set values for 605A_h (03BF_h) is shown in the table below:

Table 3- 6

Setting Value	Stop Mode
0x00	Slow down freely and keep operating freely.
0x01	Slow down on slow down ramp set in 6084h, and keep operating freely.
0x02	Slow down on slow down ramp set in 6085h, and keep operating freely.

● Halt

Bit8 of control word 6040_h (0380_h) is the pause function bit, and the pause stop is executed when bit8 changes from 0 to 1.

The pause stop mode is selected by the object dictionary 605D_h (Halt option code), and the meaning of the setting value of 605D_h (03C0_h) is shown in the following table:

Table 3- 7

Setting Value	Stop Mode
0x00	Reserved
0x01	Slow down on slow down ramp set in 6084 _h , and stay in Clamped state.
0x02	Slow down on slow down ramp set in 6085 _h , and stay in Clamped state.

● Disabled Operation

The deactivation mode is selected using the object dictionary 605C_h (03BE_h), and the set values and meanings of 605C_h (03BE_h) are shown in the table below:

Table 3- 8

Setting Value	Stop Mode
0x00	Slow down freely and keep operating freely..
0x01	Slow down on slow down ramp set in 6084 _h , and keep operating freely.

● Shutdown

The shutdown is selected via the object dictionary 605B_h (03BD_h) and the meaning of the set values for 605B_h (03BD_h) is shown in the table below:

Table 3- 9

Setting Value	Stop Mode
0x00	Slow down freely and keep operating freely.
0x01	Slow down on slow down ramp set in 6084 _h , and keep operating freely.

4 Unit Conversion

In order to adapt to applications in various industries and to facilitate users to define their respective command units, the internal unit conversion unit of this device can convert any user unit (User Unit: UU for short) into the drive's internal operating unit (Encoder Increment: Inc for short).

4.1 608F_h : Position Encoder Resolution

This object should indicate the configured encoder increment and the number of motor revolutions. The formula for calculating the position encoder resolution is given below:

$$\text{Resolution of position encoder} = \frac{\text{increment of encoder}(608F_h : 01h)}{\text{motor turns}(608F_h : 02h)}$$

The following table provides a description of the object:

Table 4- 1

Attribution	Description
Index	608F _h
Name	Resolution of position encoder
Data structure	Array
Data type	Unsigned32

The following table describes the sub-index:

Table 4- 2

Attribution	Description	
Sub-index	01 _h (0406) _h	02 _h (0408) _h
Description	Encoder increment	Motor revolutions
Accessibility	rw	rw
Mappability	NO	NO
Data type	Unsigned32	Unsigned32
Default value	10000	1

4.2 6091_h : Transmission Ratios

This object shall indicate the numbers of motor revolutions and drive-shaft revolutions. The formula to calculate Gear ratio is as follows:

$$\text{Gear ratio} = \frac{\text{Motor revolutions}(6091\text{h} : 01\text{h})}{\text{drive - shaft revolutions}(6091\text{h} : 02\text{h})}$$

The object description is specified in the table below :

Table 4- 3

Attribution	Description
Index	6091 _h (040E _h , 0410 _h)
Name	Gear ratio
Data structure	Array
Data type	Unsigned32

The sub-indexes description is specified in the table below:

Table 4- 4

Attribution	Description	
Sub-index	01 _h (040E _h)	02 _h (0410 _h)
Description	Motor revolutions	Shaft revolutions
Accessibility	rw	rw
Mappability	No	No
Data type	Unsigned32	Unsigned32
Default value	1	1

Gear ratio is used to specify the proportional relationship between the drive shaft displacement and the motor shaft displacement.

- Relationship between shaft position feedback (encoder unit) and drive shaft position feedback (instruction unit):

$$\text{Motor Shaft Position Feedback} = \text{Drive Shaft Position Feedback} \times \text{Gear Ratio}$$

- Relationship between speed of motor(rpm) and speed of shaft (command unit /s):

$$\text{speed of motor(rpm)} = \frac{\text{speed of shaft} \times \text{gear ratio (6091h)}}{\text{encoder resolutions}} \times 60$$

- Relationship between motor acceleration (rpm / ms) and shaft speed (command unit /s²):

$$\text{motor shaft acceleration} = \frac{\text{acceleration of shaft} \times \text{gear ratio(6091h)}}{\text{encoder resolutions}} \times \frac{1000}{60}$$

4.3 607E_h : Polarity

This object dictionary is used to set the position and velocity commands to be multiplied by 1 or -1. Position polarity is only used within the Contour Position and Cyclic Synchronized Position modes and has no effect on the Home Return mode. Velocity polarity is only used within the Contour Velocity Mode and Cyclic Synchronized Velocity Mode. The specific definitions are shown in the table below:

Table 4- 5

Bit-Value Meaning		
7	6	5~0
Position polarity	Velocity polarity	reserved (0)

The bits in the table above: 0 is multiplied by 1, 1 is multiplied by -1. For example, if Position polarity = 0, position demand value will be multiplied by 1. The object description is specified in the table below:

Table 4- 6

Attribution	Description
Index	607E _h (03F3 _h)
Name	Polarity
Data structure	Variable
Accessibility	rw
Data type	Unsigned8
Default value	0

4.4 Example

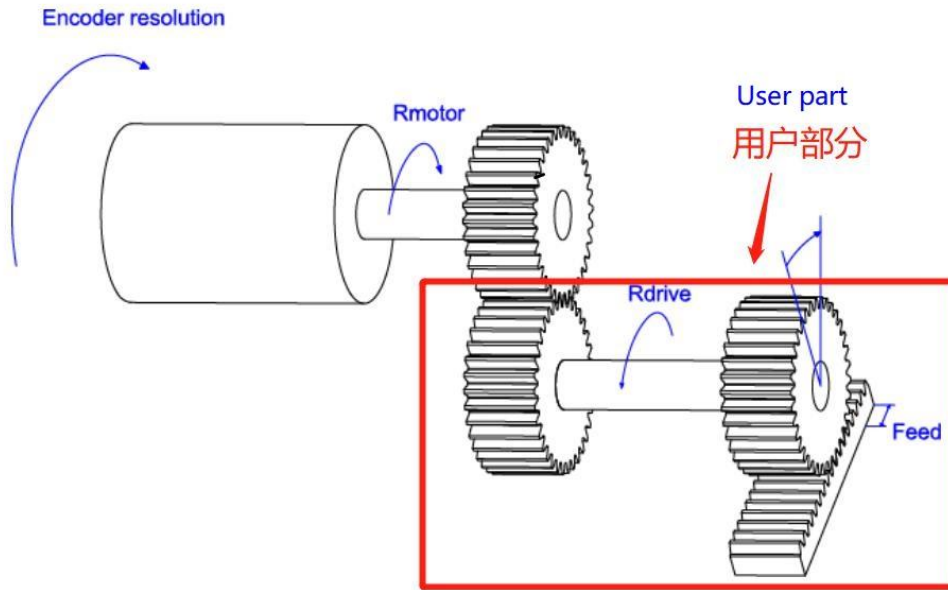


Figure 4- 1

As shown in the figure above, R_{motor} is the motor shaft and R_{drive} is the drive shaft. Suppose: the Encoder resolution = 10000 as the motor shaft rotates 1 revolution; R_{motor} rotates 3 revolutions when the R_{drive} rotates 1 revolution; from the conditions above, it can be seen that as the drive shaft rotates 1 revolution, the motor needs to operate $3 * 10000$ encoder codes, which means that the motor rotates 3 revolutions. Parameters can be calculated according to the following formulas:

Position encoder resolution 608F_h :

$$\text{Resolution of position encoder} = \frac{\text{increment of encoder}(608F_h : 01h)}{\text{motor turns } (608F_h : 02h)} = \frac{10000}{1} = 10000$$

Gear Ratio 6091_h :

$$\text{Gear ratio} = \frac{\text{Motor revolutionhs}(6091h : 01h)}{\text{drive - shaft revolutions}(6091h : 02h)} = \frac{3}{1} = 3$$

For example, under the above conditions, a target position of 10,000 pulses (user units) is converted to internal encoder units: $10000 * 3$ (transmission ratio) = 30000 pulses.